CS3281 / CS5281

Inter-Process Communications

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Overview

- Interprocess communication is about ways to make processes "talk" to one another or "synchronize" with one another
  - Recall that processes have separate virtual address spaces, so they can't just share variables
- Three big categories of IPC
  - Communication: how do processes exchange data
    - Example: send a list of files from one process to another
  - Synchronization: synchronize the actions of processes or threads
    - Think of synchronization as how to coordinate actions
      - Example: allow processes to avoid updating the same part of a file simultaneously
  - Signals: can be used for synchronization (but are primarily for other purposes)

Taxonomy

- The figure on the right shows a taxonomy of IPC mechanisms
  - We've looked at signals, mutexes, condition variables, and semaphores
  - We'll look at pipes next

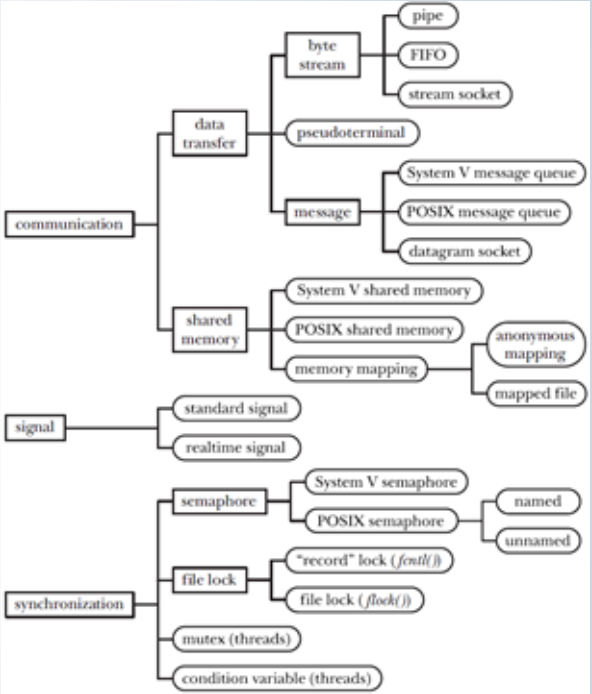


Figure 43-1: A taxonomy of UNIX IPC facilities

*Figure from *The Linux Programming Interface* by Michael Kerrisk

Fundamental Concept: File Descriptors

- File descriptor: normally small, non-negative integers that the kernel uses to identify the files accessed by a process
  - Example: when a process opens an existing file or creates a new file, the kernel returns a file descriptor that can be used to read or write the file
- All shells open three descriptors when a new program is run:
  - 0: standard input
  - 1: standard output
  - 2: standard error
- If nothing special is done: all of them are connected to the terminal
  - In other words, input comes from the terminal, and output (including errors) are written to the terminal
- How does the kernel view and use file descriptors?

Kernel Data Structures for I/O

- (a) Per-process file descriptor table contains a pointer to a file table entry
- (b) Kernel maintains file table for all open files
  - Flags are read, write, append, sync, etc
  - *i.e., one open file instance*
- (c) Kernel maintains V-node table
  - Each entry contains information about the type of file and pointers to functions that operate on the file
  - Different "read" function for files vs. folders vs. devices
  - Usually also contains the file's *i-node*, which is its metadata

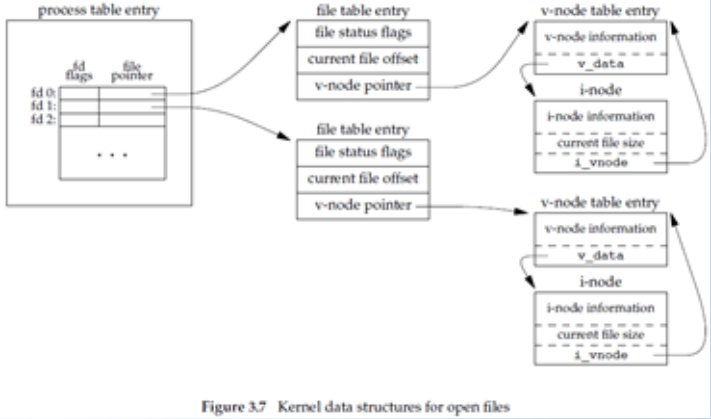


Figure 3.7 Kernel data structures for open files

*Figure from *Advanced Programming in the Unix Environment 3rd Edition* by Richard Stevens and Stephen Rago

Kernel Data Structures for I/O

- Figure on the right shows two processes with the same open file
- Two fds from the same process can also point to the same file table entry
  - The dup() system call

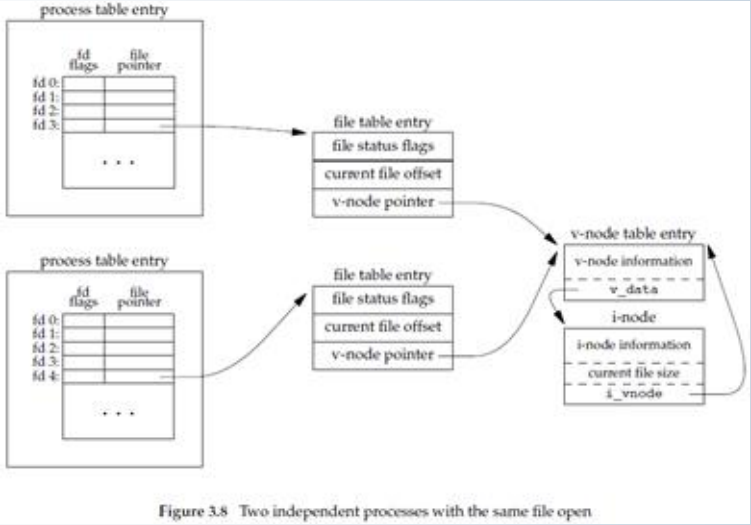


Figure 3.8 Two independent processes with the same file open

*Figure from *Advanced Programming in the Unix Environment 3rd Edition* by Richard Stevens and Stephen Rago

Kernel Data Structures for I/O

- Two fds from different processes can also point to the same file table entry
  - For instance, after a fork()
- Let's use this knowledge to do something interesting

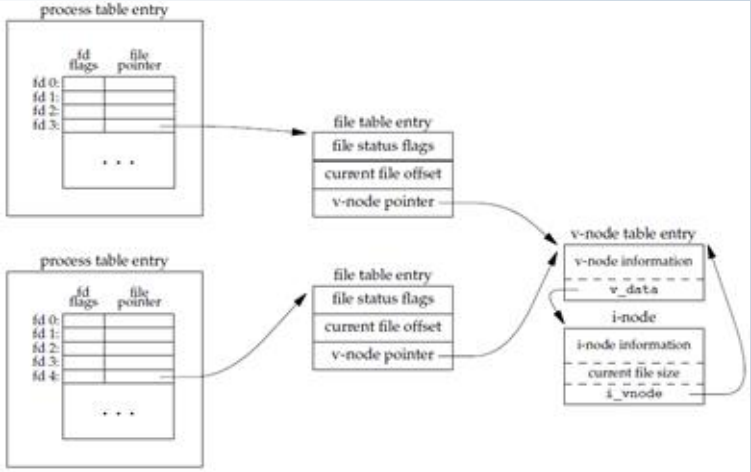


Figure 3.8 Two independent processes with the same file open

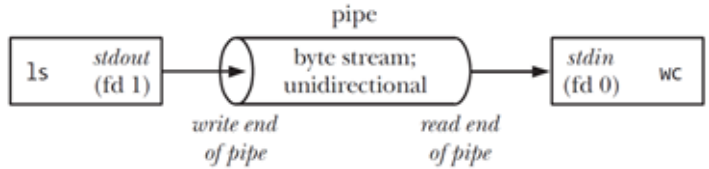
*Figure from *Advanced Programming in the Unix Environment 3rd Edition* by Richard Stevens and Stephen Rago

Pipes: Motivation

- Big “real-world” use: connecting programs
  - How can the shell send the output of one program to the input of another program?
- Example
  - The ls program will show the contents of a directory
  - The wc program will count the number of lines in its input
  - How can we use these together to count the number of files in a directory?
- One (poor) solution:
  - Run ls and send its output to a temporary file (temp.txt)
  - Run wc using temp.txt as the input
  - Delete temp.txt

Pipes

- Better solution: use a pipe!
  - Think of it as a piece of “plumbing” that lets data flow from one process to another



- More formally: a pipe is a byte-stream IPC mechanism that provides a one-way flow of data between processes
  - All data written to the pipe is routed by the kernel to another process, which can then read it
  - Think of them as open files that have no corresponding image on your filesystem

Using Pipes

- A process can create a pipe using the pipe system call:
  - `int pipe(int filedes[2]);`
  - That is, the pipe system call takes an integer array of size 2 (returns 0 on success)
    - filedes[0] can be used to read from the pipe
    - filedes[1] can be used to write to the pipe

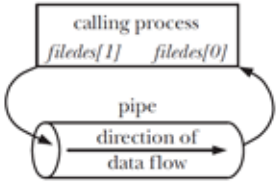


Figure 44-2: Process file descriptors after creating a pipe

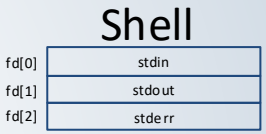
*Figure from *The Linux Programming Interface* by Michael Kerrisk

The Shell and Pipes

- Back to our motivation: how can pipes help the shell connect the output of one program to the input of another program?
  - Recall how the shell works:
    - Read a command
    - Do a fork() to create a new process
    - Do an exec() in the new process to run the program
    - Repeat
- We need the "standard output" of one process to go to the "standard input" of another process
  - Solution: have the shell "fix-up" the two processes' file descriptors!

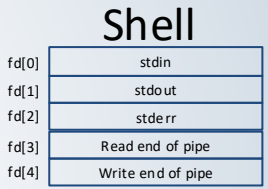
The Shell and Pipes (cont.)

- **Step 0**: the shell has the three “standard” file descriptors open



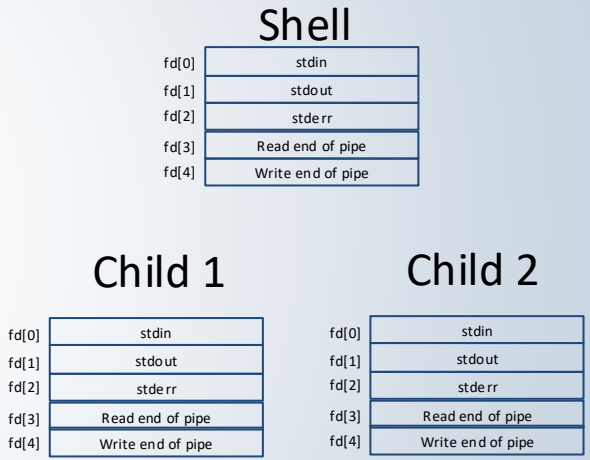
The Shell and Pipes (cont.)

- **Step 0**: the shell has the three “standard” file descriptors open
- **Step 1**: the shell process calls pipe() to create the pipe



The Shell and Pipes (cont.)

- **Step 0**: the shell has the three “standard” file descriptors open
- **Step 1**: the shell process calls pipe() to create the pipe
- **Step 2**: the shell process calls fork() twice to create the two child processes



The Shell and Pipes (cont.)

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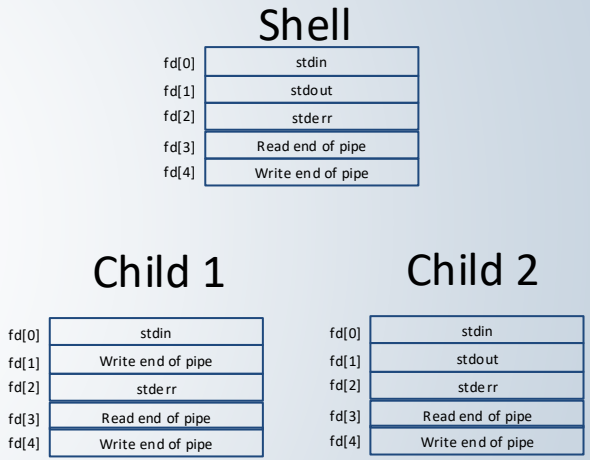
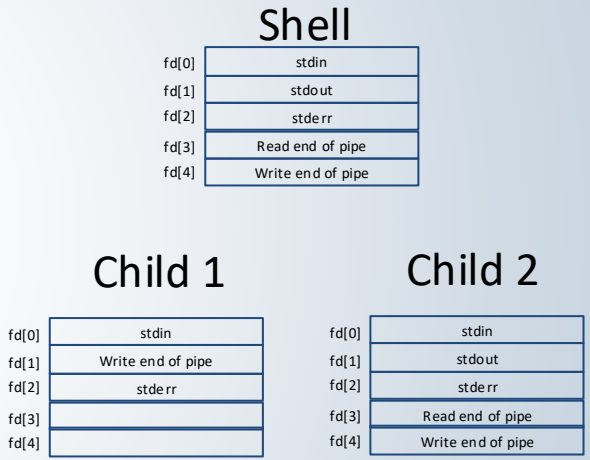
- **Step 3**: in the first child process, the write end of the pipe is dup'ed onto the file descriptor for standard output



- The dup system call:
  - `int dup2(int oldfd, int newfd);`
  - Duplicates the descriptor in oldfd to the descriptor in newfd

The Shell and Pipes (cont.)

- **Step 0**: the shell has the three "standard" file descriptors open
- **Step 1**: the shell process calls pipe() to create the pipe
- **Step 2**: the shell process calls fork() twice to create the two child processes
- **Step 3**: in the first child process, the write end of the pipe is dup'ed onto the file descriptor for standard output
  - Child process closes both pipe fds and calls exec



The Shell and Pipes (cont.)

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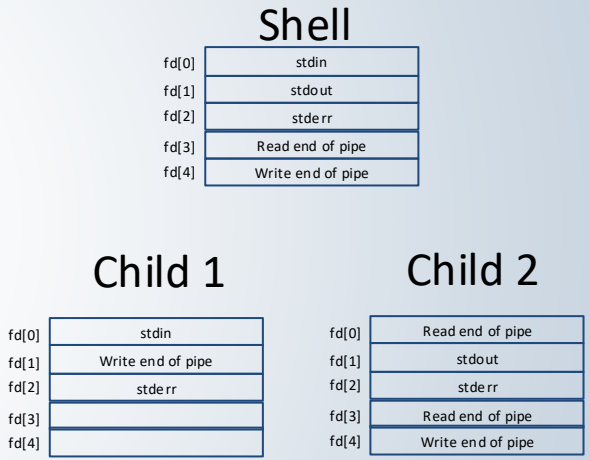
- **Step 4**: in the second child process, the read end of the pipe is dup'ed onto the file descriptor for standard input



The Shell and Pipes (cont.)

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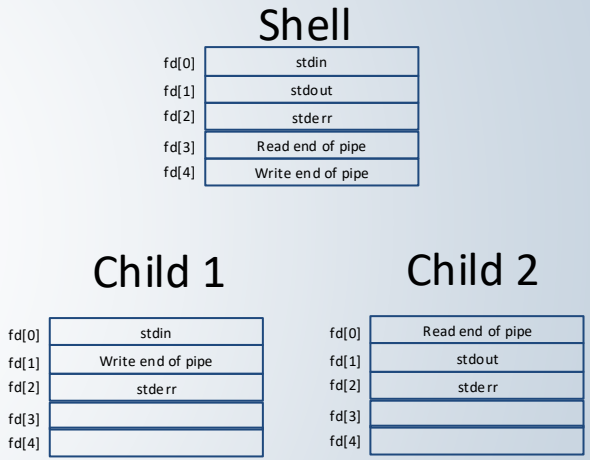

The Shell and Pipes (cont.)

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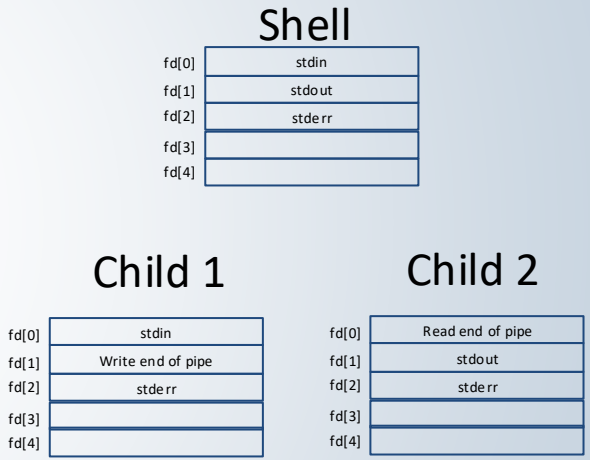
  - Child process closes both pipe fds and calls exec
- **Step 4**: in the second child process, the read end of the pipe is dup'ed onto the file descriptor for standard input
  - Child process closes both pipe fds and calls exec
- **Step 5**: shell process closes both pipe fds



Other IPC Mechanisms (optional readings)

Other types of IPC

- Message queues let processes exchange data in the form of messages
  - Contrast to pipes, which are streams of data
- POSIX shared memory lets us share a mapped region between unrelated processes without needing to create a corresponding mapped file

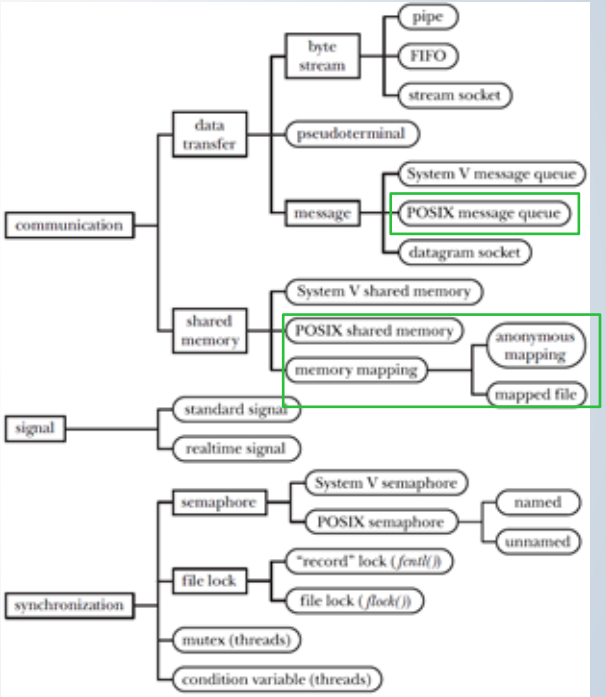


Figure 43-1: A taxonomy of UNIX IPC facilities

*Figure from *The Linux Programming Interface* by Michael Kerrisk

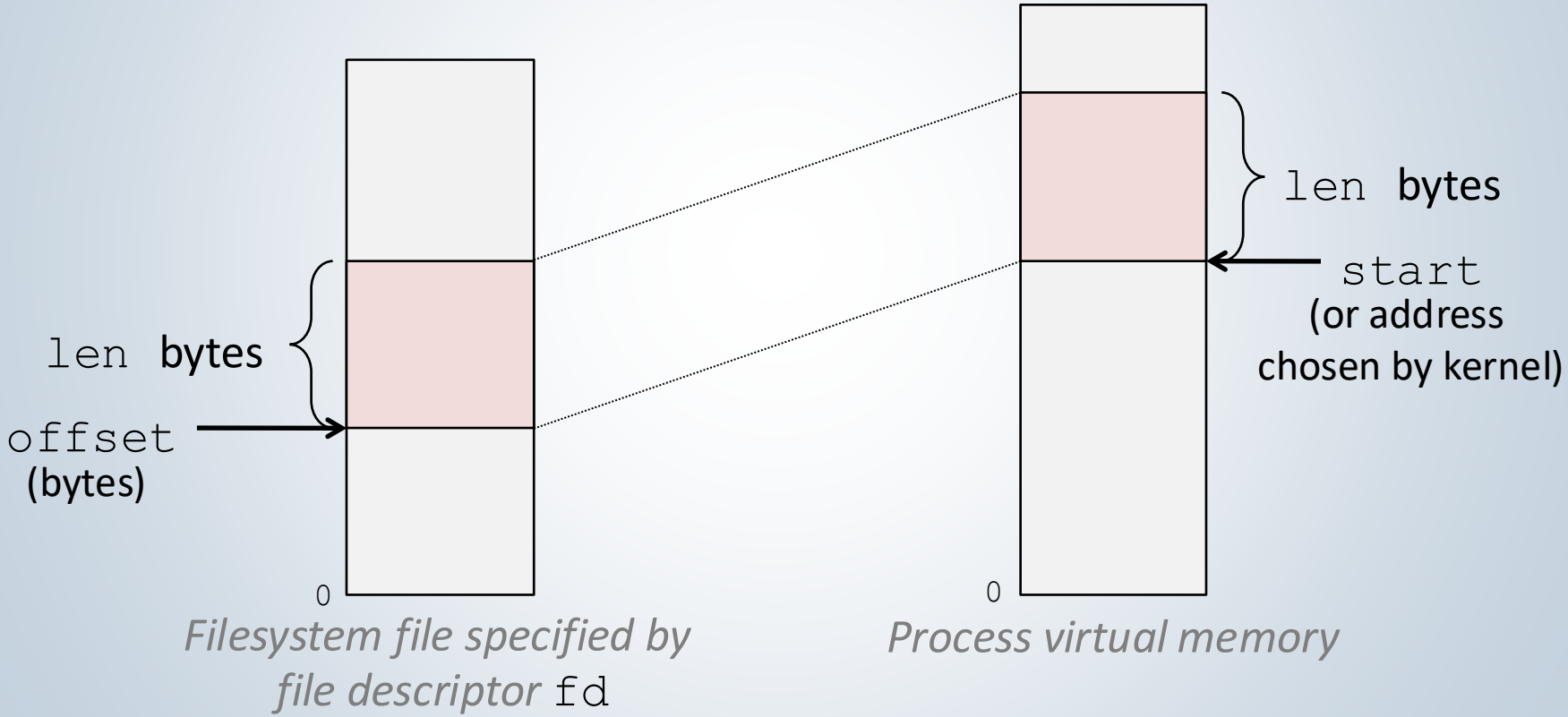
User-Level Memory Mapping (Linux)

```
void *mmap(void *start, int len,
           int prot, int flags, int fd, int offset)
```

- Map **`len`** bytes starting at offset **`offset`** of the file specified by file description **`fd`**, preferably at address **`start`**
  - **`start`**: may be 0 for "pick an address"
  - **`prot`**: PROT_READ, PROT_WRITE, ...
  - **`flags`**: MAP_ANON, MAP_PRIVATE, MAP_SHARED, ...
- Return a pointer to start of mapped area (might not be **`start`**)

User-Level Memory Mapping

```
void *mmap(void *start, int len, int prot, int flags, int fd, int offset)
```



Shared Memory

- Can use mmap to create a shared memory region
  - Only accessible to future children processes
- Can also use mmap to map a region of memory to a file that may not be stored on disk
- By “naming” that memory as a file in the filesystem, other processes can map it into their own address space
- mmap can therefore be used to share memory between processes, and allow for shared-memory based IPC
  - Have to worry about shared-memory synchronization!

Message queues

- Allow processes to exchange data in the form of *messages*
  - This means the reader receives whole messages as written by the writer
- Contrast this to pipes
  - The receiver gets a stream of bytes; they might read one byte or many.
- Five main functions for using POSIX message queues:
  - mq_open(): creates a new mq or opens an existing queue
  - mq_send(): writes message to queue
  - mq_receive(): reads a message from a queue
  - mq_close(): closes a message queue
  - mq_unlink(): removes queue name and marks it for deletion when all processes have closed it

Opening a message queue

- mqd_t mq_open(const char *name, int oflag, mode_t mode, struct mq_attr *attr)
  - Values for oflag and mode are shown in tables 52-1 and 15-4
  - struct mq_attr shown below

Table 15-4: Constants for file permission bits

Constant	Octal value	Permission bit
S_ISUID	04000	Set-user-ID
S_ISGID	02000	Set-group-ID
S_ISVTX	01000	Sticky
S_IRUSR	0400	User-read
S_IWUSR	0200	User-write
S_IXUSR	0100	User-execute
S_IRGRP	040	Group-read
S_IWGRP	020	Group-write
S_IXGRP	010	Group-execute
S_IROTH	04	Other-read
S_IWOTH	02	Other-write
S_IXOTH	01	Other-execute

Table 52-1: Bit values for the *mq_open() oflag* argument

Flag	Description
O_CREAT	Create queue if it doesn't already exist
O_EXCL	With O_CREAT, create queue exclusively
O_RDONLY	Open for reading only
O_WRONLY	Open for writing only
O_RDWR	Open for reading and writing
O_NONBLOCK	Open in nonblocking mode

```
struct mq_attr {
    long mq_flags;       /* Message queue description flags: 0 or
                            O_NONBLOCK [mq_getattr(), mq_setattr()] */
    long mq_maxmsg;      /* Maximum number of messages on queue
                            [mq_open(), mq_getattr()] */
    long mq_msgsize;     /* Maximum message size (in bytes)
                            [mq_open(), mq_getattr()] */
    long mq_curmsgs;     /* Number of messages currently in queue
                            [mq_getattr()] */
};
```

*Figure from *The Linux Programming Interface* by Michael Kerrisk

Message queue descriptors

- A message queue descriptor is per-process
  - Similar to file-descripto

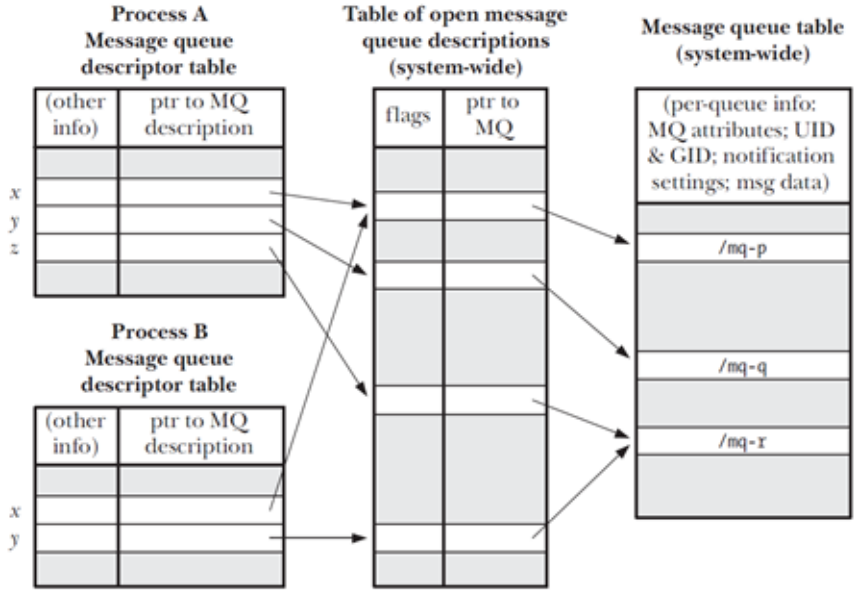


Figure 52-1: Relationship between kernel data structures for POSIX message queues